

Publications (107)

Scientific publications on research performed at the Alpine Research and Education Station (ALPFOR) Furkapass, Switzerland 2440 m.

The below list contains all publications that contain data collected at ALPFOR.

Publications that emerged from ALPFOR activities (reviews, synthesis papers), international networking closely associated with ALPFOR (e.g., GMBA-DIVERSITAS), or which became reality due to grants acquired by the ALPFOR-team (e.g. SNF VALUrsern) and relate to themes initiated by ALPFOR but were performed by other research groups are labelled with an * at the end of the reference (17 publications). The list (ordered alphabetically) includes early research performed at Furka pass, before ALPFOR gained formal responsibility of the station buildings.

- Ægisdóttir HH, Kuss P, Stöcklin J (2009) Isolated populations of a rare alpine plant show high genetic diversity and considerable population differentiation. *Annals of Botany* 104:1313-2009
- Alaoui A, Willmann W, Jasper K, Felder G, Herger F, Magnusson J, Weingartner R (2014) Modelling the effects of land use and climate changes on hydrology in the Ursern Valley, Switzerland *Hydrological Processes* 28:3602–3614 *
- Blumer P, Diemer M (1996) The occurrence and consequences of grasshopper herbivory in an alpine grassland, Swiss Central Alps. *Arctic Alp Res* 28:435-440
- Bramer I, Anderson BJ, Bennie J, Bladon AJ, De Frenne P, Hemming D, Hill RA, Kearney MR, Körner C, Korstjens AH, Lenoir J, Maclean IMD, Marsh CD, Morecroft MD; Ohlemüller R, Slater HD, Suggitt AJ, Zellweger F, Gillingham PK (2018) Advances in monitoring and modelling climate at ecologically relevant scales. *Adv Ecol Res* 58:101- 161
- Budge K, Leifeld J, Hiltbrunner E, Fuhrer J. 2010. Litter quality and pH are strong drivers of carbon turnover and distribution in alpine grassland soils. *Biogeosciences Discuss.* 7: 6207–6242
- Bühlmann T, Caprez R, Hiltbrunner E, Körner C, Niklaus PA (2017) Nitrogen fixation by *Alnus* species boosts soil nitrous oxide emissions. *Eur J Soil Sci* 68:740-748
- Bühlmann T, Hiltbrunner E, Körner C (2013) Die Verbuschung des Alpenraums durch die Grünerle. FactSheet, Schweizerische Akademie der Wissenschaften
- Bühlmann T, Hiltbrunner E, Körner C (2014) *Alnus viridis* expansion contributes to excess reactive nitrogen release, reduces biodiversity and constrains forest succession in the Alps. *Alp Bot* 124: 187-191.
- Bühlmann T, Hiltbrunner E, Körner C, Rihm B, Achermann B (2015) Induction of indirect N₂O and NO emissions by atmospheric nitrogen deposition in (semi-) natural ecosystems in Switzerland. *Atmos Environ* 103:94-101 *
- Bühlmann T, Körner C, Hiltbrunner (2016) Shrub expansion of *Alnus viridis* drives former montane grassland into nitrogen saturation. *Ecosystems*, accepted
- Caprez R, Niklaus PA, Körner C (2012) Forest soil respiration reflects plant productivity across a temperature gradient in the Alps. *Oecologia* 170:1143-1154 *
- Caprez R, Spehn E, Nakhutsrishvili G, Körner C (2012) Drought at erosion edges selects for a 'hidden' keystone species. *Plant Ecol Divers* 4: 303-311 *
- De Boeck H J, Hiltbrunner E, Verlinden M, Bassin S, Zeiter, M (2018) Legacy effects of climate extremes in alpine grassland. *Frontiers in Plant Science* 9:1586.

- De Boeck HJ, Bassin S, Verlinden M, Zeiter M, Hiltbrunner E (2015) Simulated heat waves affected alpine grassland only in combination with drought. *New Phytol* doi: 10.1111/nph.13601
- De Boeck HJ, Hiltbrunner E, Jentsch A and Vandvik V (2019) Editorial: Responses to climate change in the cold biomes. *Front. Plant Sci.* 10:347.
- De Witte LC, Armbruster GFJ, Gielly L, Taberlet P, Stöcklin J (2012) AFLP markers reveal high clonal diversity and extreme longevity in four key arctic-alpine species. *Mol Ecol* 21:1081-1097
- De Witte LC, Scherrer D, Stöcklin J (2011) Genet longevity and population age structure of the clonal pioneer species *Geum reptans* based on demographic field data and projection matrix modelling. *Preslia* 83:371-386
- De Witte LC, Stöcklin J (2010) Longevity of clonal plants: why it matters and how to measure it. *Ann Bot* 106:859-870
- De Witte LC, Stöcklin J (2011) Horizontal growth in arctic-alpine clonal plants is not affected by climatic variability among regions. *Plant Ecology and Diversity* 4:329-340
- Diemer M, Körner C (1996) Lifetime leaf carbon balances of herbaceous perennial plants from low and high altitudes in the central Alps. *Funct Ecol* 10:33-43
- Diemer M, Körner C (1998) Transient enhancement of carbon uptake in an alpine grassland ecosystem under elevated CO₂. *Arctic Alp Res* 30:381-387
- Dietrich L, Körner C (2014) Thermal imaging reveals massive heat accumulation in flowers across a broad spectrum of alpine taxa. *Alp Bot* 124:27-35
- Frei ES, Scheepens JF, Stöcklin J (2012) High genetic differentiation in population of the rare alpine plant species *Campanula thyrsoidea* on a small mountain. *Alpine Botany* 122:23-34
- Heer C, Körner C (2002) High elevation pioneer plants are sensitive to mineral nutrient addition. *Basic Appl Ecol* 3:39-47
- Hefel C, Stöcklin J (2010) Flora der Furka. *Bauhinia* 22:33-59.
- Hiltbrunner E, Aerts R, Bühlmann T, Huss-Danell K, Magnusson B, Myrold DD, Reed SC, Sigurdsson BD, Körner C (2014) Ecological consequences of the expansion of N₂-fixing plants in cold biomes. *Oecologia* 176:11-24
- Hiltbrunner E, Arnaiz J, Körner C (2021) Biomass allocation and seasonal non-structural carbohydrate dynamics do not explain the success of tall forbs in short alpine grassland. *Oecologia* 197:1063-1077
- Hiltbrunner E, Körner C (2004) Sheep grazing in the high alpine under global change. In: Lüscher A, Jeangros B, Kessler W, Huguenin O, Lobsiger M, Millar N, Suter D (eds) Land use systems in grassland dominated regions. *Grassland Sciences in Europe* 9: 305-307
- Hiltbrunner E, Körner C (2018) Hotspot Furka. Biologische Vielfalt im Gebirge. Alpine Forschungs- und Ausbildungsstation Furka und Universität Basel. 60S. ISBN 978-3-033-06701-1
- Hiltbrunner E, Körner C, Meier R, Braun S, Kahmen A (2019) Data do not support large scale oligotrophication of terrestrial ecosystems. *Nature Ecol Evol*
- Hiltbrunner E, Lüscher E, van den Bergh, Körner C (2010) Ecological and hydrological consequences of green alder expansion in the Swiss Alps. In Spehn E M, Rudmann-Maurer K, Körner C (eds.). Mountain biodiversity and global change. GMBA DIVERSITAS, Basel

- Hiltbrunner E, Schwikowski M, Körner C (2005) Inorganic nitrogen storage in alpine snow pack in the Central Alps (Switzerland). *Atmos Environ* 39:2249-2259
- Hiltbrunner E, Zehnder T (2010) Die Verbuschung des Schweizerischen Alpenraumes. Inside, Konferenz der Beauftragten für Natur- und Landschaftsschutz 4/2010:22-27
- Hiltbrunner E. (2019) Mit Engadinerschafem gegen die Verbuschung des Alpenraumes. *Schweizer Zeitschrift für Ernährungsmedizin* 4:12-13
- Huck C, Körner C, Hiltbrunner E (2013) Plant species dominance shifts across erosion edge-meadow transects in the Swiss Alps. *Oecologia* 171:693-703
- Inauen N, Körner C, Hiltbrunner E (2012) No growth stimulation by CO₂ enrichment in alpine glacier forefield plants. *Glob Change Biol* 18:985-999
- Inauen N, Körner K, Hiltbrunner E (2013) Hydrological consequences of declining land use and elevated CO₂ in alpine grassland. *J Ecol* 101:86-96
- Körner C (2000) The alpine life zone under global change. *Gayana Bot (Chile)* 57(1):1-17
- Körner C (2003) *Alpine plant life*, 2nd edition, Springer, Heidelberg
- Körner C (2004) Mountain biodiversity, its causes and function. *Ambio Special report* 13:11-17 *
- Körner C (2007) Alpine ecosystems. *Encyclopedia of Life Sciences*, John Wiley, pp 1-6*
- Körner C (2018) Comparative, long-term ecosystem monitoring across the Alps: Austrian Hohe Tauern National Park, South-Tyrol and Swiss central Alps. In: Bauch K (ed) *Proc Symp Res Protected Areas. Salzburger Nationalparkfonds, Mittersill, and Austrian Acad Sci, Vienna* p. 331-337
- Körner C (2018) Concepts in empirical plant ecology. *Plant Ecol Divers* 11(4):405-428*
- Körner C (2019) Climatic controls of the global high elevation treelines. Reference Module in Earth Systems and Environmental Sciences doi: 10.1016/B978-0-12-409548-9.11998-0 *
- Körner C (2019) Plant adaptations to alpine environments. Reference Module in Earth Systems and Environmental Sciences doi 10.1016/B978-0-12-409548-9.11793-2*
- Körner C (2020) Experiments by nature – strength in realism In: Burt TP, Thompson DBA (eds.) *Curious about nature: a passion for fieldwork*. Cambridge University Press, Cambridge, Chap 27 *
- Körner C (2023) Concepts in alpine plant life. *Plants*, in press
- Körner C, Buchmann N, Bugmann H, Duelli P, Hiltbrunner E, Müller-Ferch G, Müller JP, Wildi O, Zweifel R (2007) Landökosysteme. In: ProClim O (ed) *Klimaänderungen und die Schweiz 2050. Erwartete Auswirkungen auf die Umwelt, Gesellschaft und Wirtschaft*. OcCC ProClim, Bern p. 25-40 *
- Körner C, Diemer M (1994) Evidence that plants from high altitudes retain their greater photosynthetic efficiency under elevated CO₂. *Funct Ecol* 8:58-68
- Körner C, Diemer M, Schächli B (1998) Alpenrasen unter erhöhtem CO₂. In: Körner C, Hättenschwiler S (eds) *Die Alpen und das CO₂-Problem*. Hochschulverlag AG an der ETH Zürich p. 31-64
- Körner C, Diemer M, Schächli B, Niklaus P, Arnone J (1997) The responses of alpine grassland to four seasons of CO₂ enrichment: a synthesis. *Acta Oecologica* 18:165-175
- Körner C, Diemer M, Schächli B, Zimmermann L (1996) Responses of alpine vegetation to elevated CO₂. In: Koch GW, Mooney HA (eds) *Carbon dioxide and terrestrial ecosystems*. Academic Press, San Diego p. 177-196
- Körner C, Donoghue M, Fabbro T, Hauser C, Nogues-Bravo D, Arroyo MTK, Soberon J, Speers L, Spehn EM, Sun H, Tribsch A, Tykarski P, Zbinden N (2007) Creative use of

- mountain biodiversity databases: The Kazbegi research agenda of GMBA-DIVERSITAS. *Mt Res Dev* 27:276-281 *
- Körner C, Hiltbrunner E (2018) The 90 ways to describe plant temperature. *Perspect Plant Ecol Evol Syst* 30:16-21
- Körner C, Hiltbrunner E (2021) Why is the alpine flora comparatively robust against climatic warming? *Diversity* 13:383 doi: 10.3390/d13080383
- Körner C, Jetz W, Paulsen J, Payne D, Rudmann-Maurer K, Spehn EM (2017) A global inventory of mountains for bio-geographical applications. *Alp Bot* 127:1-15 *
- Körner C, Nakhutsrishvili G, Spehn E (2006) High elevation land use, biodiversity, and ecosystem functioning. In: Spehn EM, Körner C (eds) *Land use change and mountain biodiversity*. CRC Publishers, Boca Raton, pp 3-21 *
- Körner C, Ohsawa M (2006) Mountain systems. In: Hassan R, Scholes R, Ash N (eds) *Ecosystem and human well-being: current state and trends*, vol. 1, Millennium Ecosystem Assessment, Island Press, Washington, pp 681-716 *
- Körner C, Spehn EM, Nakhutsrishvili G (2004) Grazing the highlands: food, biodiversity and catchment implications. In: Lüscher A, Jeangros B, Kessler W, Huguenin O, Lobsiger M, Millar N, Suter D (eds) *Land use systems in grassland dominated regions*. vdf Hochschulverlag Ag, ETH Zürich p. 278-280
- Körner C, Paulsen J (2017) A geostatistical and bioclimatological comparison of the Central Great Caucasus and the Central Alps. In: Nakhutsrishvili et al (eds) *Plant Diversity in the Central Great Caucasus: A Quantitative Assessment*. Springer International Publishing, Switzerland p. 1-9;161-170 *
- Körner C, Paulsen J, Pelaez-Riedl S (2003) A bioclimatic characterisation of Europe's alpine areas. In: Nagy L, Grabherr G, Körner C, Thompson DBA (eds) *Alpine biodiversity in Europe*. *Ecol Studies* 167:13-28, Springer, Berlin
- Körner C, Paulsen J, Spehn EM (2011) A definition of mountains and their bioclimatic belts for global comparison of biodiversity data. *Alp Botany* 121:73-78
- Körner C, Riedl S, Keplinger T, Richter A, Wiesenbauer J, Schweingruber F, Hiltbrunner E (2019) Life at 0 °C: the biology of the alpine snowbed plant *Soldanella pusilla*. *Alp Botany* doi: 10.1007/s00035-019-00220-8
- Körner C, Spehn E (2019) A Humboldtian view of mountains. *Science* 365(6458):1061-1061 *
- Körner C, Berninger UG, Daim A, Eberl T, Fernandez Mendoza F, Fureder L, Grube M, Hainzer E, Kaiser R, Meyer E, Newesely C, Niedrist G, Niedrist GH, Petermann JS, Seeber J, Tappeiner U, Wickham S (2022) Long-term monitoring of high-elevation terrestrial and aquatic ecosystems in the Alps – a five-year synthesis. *Eco Mont* 14:48-69
- Körner C, Hoch G, (2023) Not every high-latitude or high-elevation forest edge is a treeline. *J Biogeography*, DOI: 10.1111/jbi.14593
- Körner C, Fajardo A, Hiltbrunner E (2023) Biogeographic implications of plant stature and microclimate in cold regions. *Comms Biol*, <https://doi.org/10.1038/s42003-022-03732-y>
- Körner C, Möhl P, Hiltbrunner E (2023) Four ways to define the growing season. *Ecology Letters*, DOI: 10.1111/ele.14260
- Kosonen Z, Schnyder E, Hiltbrunner E, Thimonier A, Schmitt M, Seitler E, Thöni L. (2019). Current atmospheric nitrogen deposition still exceeds critical loads for sensitive, semi-natural ecosystems in Switzerland. *Atmospheric Environment* 211:214-225.

- Kuss P, Armbruster GFJ, Aegisdóttir HH, Scheepens JF, Stöcklin J (2011) Spatial genetic structure of *Campanula thyrsoidea* across the European Alps: Indication for glaciation-driven allopatric subspeciation. *Perspect Plant Ecol Evol Syst* 13:101-110
- Kuss P, Pluess AR, Aegisdóttir HH, Stöcklin J (2008) Spatial isolation and genetic differentiation in naturally fragmented plant populations of the Swiss Alps. *J Plant Ecol* 1:149-159
- Merz A, Alewell C, Hiltbrunner E, Bänninger D (2009) Plant-compositional effects on surface runoff and sediment yield in subalpine grassland *Journal of Plant Nutrition and Soil Science* 172: 777–788
- Möhl P, Hiltbrunner E, Körner C (2020) Halving sunlight reveals no carbon limitation of aboveground biomass production in alpine grassland. *Glob Change Biol* 26:1857–1872
- Möhl P, Von Büren R, Hiltbrunner E (2022) Growth of alpine grassland will start and stop earlier under climate warming. *Nature Communications* 13:7398.
- Möhl P, Vorkauf M, Kahmen A, Hiltbrunner E (2023) Recurrent summer drought affects biomass production and community composition independently of snowmelt manipulation in alpine grassland. *Journal of Ecology* (in press)
- Monteiro A, Fava F, Hiltbrunner E, Bocchi S. 2011. Mapping of land cover changes and GIS-based logistic regression on spatial variables revealed considerable loss of permanent meadows in the lowlands of Italian Valtellina valley. *Journal Landscape and Urban Planning* 100: 287-294
- Müller MH, Alaoui A, Alewell C (2016) Water and solute dynamics during rainfall events in headwater catchments in the Central Swiss Alps under the influence of green alder shrubs and wetland soils, *Ecohydrology*, DOI: 10.1002/eco.1692 *
- Müller MH, Weingartner R, Alewell C (2013) Importance of vegetation, topography and flow paths for water transit times of base flow in alpine headwater catchments. *Hydrol Earth Syst Sci* 17:1661-1679 *
- Nagelmüller S, Hiltbrunner E, Körner C (2015) Critically low soil temperatures for root growth and root morphology in three alpine plant species. *Alp Bot*. DOI: 10.1007/s00035-015-0153-3
- Nagelmüller S, Hiltbrunner E, Körner C (2017) Low temperature limits for root growth in alpine species are set by cell differentiation. *Aob Plants* 9:plx054; doi:10.1093/aobpla/plx054
- Newesely C, Tappeiner U (2018) Climatological reference data of a newly established long-term monitoring program in the central Alps. In: Bauch K (ed) *Proc Symp Res Protected Areas*. Salzburger Nationalparkfonds, Mittersill, and Austrian Acad Sci, Vienna p. 449-451
- Obojes N, Bahn M, Tasser E, Walde J, Inauen N, Hiltbrunner E, Saccone P, Locht J, Clement JC, Lavorel S, Tappeiner U, Körner C (2015) Vegetation effects on the water balance of mountain grasslands depend on climatic conditions. *Ecohydrol* 8:552-569
- Sage RF, Schappi B, Körner C (1997) Effect of atmospheric CO₂ enrichment on Rubisco content in herbaceous species from high and low altitude. *Acta Oecologica* 18:183-192
- Schappi B, Körner C (1996) Growth responses of an alpine grassland to elevated CO₂. *Oecologia* 105:43-52
- Schappi B, Körner C (1997) In situ effects of elevated CO₂ on the carbon and nitrogen status of alpine plants. *Functional Ecology* 11:290-299

- Scheepens JF, Frei ES, Armbruster GFJ, Stöcklin J (2012) Pollen dispersal within and into a population of the alpine monocarpic plant *Campanula thyrsoidea*. *Annals of Botany* 110:1489-1501
- Scherrer D, Körner C (2010) Infra-red thermometry of alpine landscapes challenges climatic warming projections. *Glob Change Biol* 16:2602-2613
- Scherrer D, Körner C (2011) Topographically controlled thermal-habitat differentiation buffers alpine plant diversity against climate warming. *J Biogeogr* 38:406-416
- Scherrer D, Schmid S, Körner C 2010: Elevational species shifts in a warmer climate are overestimated when based on weather station data. *Intern J of Biometeorology* 55:645-654
- Schmid S, Hiltbrunner E, Spehn E M, Lüscher A, Scherer-Lorenzen M. 2011. Impact of experimentally induced summer drought on biomass production in alpine grassland in: Pötsch EM Krautzer B Hopkins A (eds.) *Grassland Farming and Land Management Systems in Mountainous Regions, Grassland Science in Europe Vol 16*, pp. 214-216
- Scholz K, Hammerle A, Hiltbrunner E, Wohlfahrt G (2018) Analyzing the Effects of Growing Season Length on the Net Ecosystem Production of an Alpine Grassland Using Model-Data Fusion. *Ecosystems* 21:982-999
- Seeber J, Newesely C, Steinwandter M, Rief A, Körner C, Tappeiner U, Meyer E (2021) Soil invertebrate abundance, diversity, and community composition across steep high elevation snowmelt gradients in the European Alps. *Arct Antarct Alp Res* 53:288-299
- Spehn E, Körner C (2017) Climate change impacts on alpine nature. *Natur und Landschaft* 92:407-411 *
- Spehn EM, Körner C (2005) A global assesment of mountain biodiversity and its function. In: Huber UM, Bugmann KM, Reasoner MA (eds): *Global change and mountain regions*. Springer, Dordrecht p. 393-400 *
- Spehn EM, Liberman M, Körner C (2006): *Land Use Change and Mountain Diversity*. CRC Press, Boca Raton *
- Steiner BL, Armbruster GFJ, Scheepens JF, Stöcklin J (2012) Distribution of bulbil- and seed producing plants of *Poa alpina* (Poaceae) and their growth and reproduction in common gardens suggest adaptation to different elevations. *Am J Bot* 99:2035-2044
- Stöcklin J, Kuss P, Pluess AR (2009) Genetic diversity, phenotypic variation and local adapataion in the alpine landscape: case studies with alpine plant species. *Botanica Helvetica* 119:125-133
- Tackenberg O, Stöcklin J (2008) Wind dispersal of alpine plant species: A comparison with lowland species. *J Vegetation Science* 19:109-118
- Tiusanen M et al, Körner C (2019) Flower-visitor communities of an arcto-alpine plant—Global patterns in species richness, phylogenetic diversity and ecological functioning. *Mol Ecol* 28:318-335
- Van den Bergh T, Inauen N, Hiltbrunner E, Körner C (2013) Climate and plant cover co-determine the elevational reduction in evapotranspiration in the Swiss Alps. *J Hydrol* 500:75-83
- van den Bergh T, Körner C, Hiltbrunner E (2018) *Alnus* shrub expansion increases evapotranspiration in the Swiss Alps. *Reg Environ Change* 18:1375-1385
- Von Büren RS, Hiltbrunner E (2022) Low winter temperatures and divergent freezing resistance set the cold range limit of widespread alpine graminoids. *J Biogeogr* 49:1562-1575

- Vorkauf M, Kahmen A, Körner C, Hiltbrunner E (2021) Flowering phenology in alpine grassland strongly responds to shifts in snowmelt but weakly to summer drought. *Alp Bot* 131:73-88
- Vorkauf M, Marty C, Kahmen A, Hiltbrunner E (2021) Past and future snowmelt trends in the Swiss Alps: the role of temperature and snowpack. *Clim Change* 165:44 doi: 10.1007/s10584-021-03027-x
- Vorkauf M, Steiger R, Abegg B, Hiltbrunner E (2022) Snowmaking in a warmer climate: an in-depth analysis of future water demands for the ski resort Andermatt-Sedrun-Disentis (Switzerland) in the twenty-first century. *Int J Biometeorol* 1–15.
- Yang Y, Siegwolf RTW, Körner C (2015) Species specific and environment induced variation of delta-13C and delta-15N in alpine plants. *Front Plant Sci* 6: 423.
- Zhu Y, Siegwolf RTW, Durka W, Körner C (2010) Phylogenetically balanced evidence for structural and carbon isotope responses in plants along elevational gradients. *Oecologia* 162:853-863

Books strongly influenced by, and containing data from ALPFOR/Furkapass (6)

- Chapin FS III, Körner C (1995) Arctic and alpine biodiversity: Patterns, causes and ecosystem consequences. *Ecol Studies* 113, Springer, Berlin
- Körner C, Spehn EM (2002) Mountain biodiversity, a global assessment. The Parthenon Publishing Group, Boca Raton p. 3-20
- Körner C (2003, 2021) *Alpine Plant Life* (2nd and 3rd ed.), Springer, Berlin (Chinese edition, by Science Publishers, Beijing; Edition in Georgian language, Tbilisi, Georgia funded by DEZA)
- Körner C (2012) *Alpine Treelines*, Springer, Basel
- Körner C (2002, 2007, 2014) *Vegetation der Erde*. In: Strasburger, *Lehrbuch der Botanik für Hochschulen* Fischer/Spektrum/Springer, Heidelberg-Berlin
- Spehn EM, Liberman M, Körner C (2006): *Land Use Change and Mountain Diversity*. CRC Press, Boca Raton